

Floote Flood Reporting & SOS System

Smart Routing IoT System — Thesis Defense–Ready Executive Report (Short Version)

1) Project Summary (Concise)

Floote is a mobile disaster-response app with two core functions, integrated with a **Smart Routing IoT System** that combines citizen flood data, optional IoT water-level telemetry, and cloud routing services to steer users and rescuers away from impassable roads.

SOS Dispatch Workflow

- User sends emergency SOS request.
- Request is tracked through statuses (`submitted`, `received`, `dispatching`, `en\_route`, `closed`).
- Rescuer receives and navigates to user.

Incident Flood Reporting Workflow

- User submits flood reports with coordinates and optional photo.
- System prevents duplicate reports in recently reported nearby areas.
- Reports are used as **hazard inputs** for safe routing logic inside the smart routing pipeline.

Smart Routing IoT System (concise)

- **Inputs:** `user\_reports` (crowdsourced hazards), future IoT sensor streams (e.g. water depth), and road network data via Google Routes/Roads (through FastAPI).
- **Processing:** FastAPI evaluates flood-aware routes; the mobile app uses Google Navigation SDK for live guidance; reroute policy favors stability unless hazards or position materially change.
- **Output:** Safer suggested paths for citizens and rescuers, aligned with verified or pending hazard decisions.

The design prioritizes:

- **Safety** — avoid hazardous roads.
- **Data quality** — location normalization + duplicate control.
- **Operational clarity** — status-driven rescue lifecycle.

2) Layman's Explanation (Short)

Think of Floote like this:

- If someone is in danger, they press **SOS** and rescuers get the request.
- If someone sees flooding, they submit a **flood report** with location.
- The app checks if that area was already reported recently to avoid duplicates.

- **Smart routing** uses those reports (and, when deployed, IoT sensors) so navigation avoids dangerous roads and guides users and rescuers more safely.

3) Current System Architecture

High-level components:

- **Mobile App (Flutter)** — map, SOS, incident report tab, navigation.
- **Supabase** — auth, `user\_reports`, `sos\_dispatches`, `sos\_dispatch\_offers`, `profiles`, RPCs and triggers.
- **FastAPI Backend** — flood-aware routing (`/route` and related services), hazard-aware path computation.
- **Google Routes / Roads APIs** — road-following, route geometry, hazard deflection (via backend keys).
- **Google Navigation SDK** — turn-by-turn guidance on device.
- **Google Places / Geocoding (web-style calls)** — optional address and landmark resolution for reports.

```

flowchart LR
    A[Mobile App - Flutter] --> B[Supabase]
    A --> C[FastAPI Smart Routing]
    C --> D[Google Routes/Roads APIs]
    A --> E[Google Navigation SDK]
    A --> F[Google Places/Geocode Web APIs]
    C --> G[Smart Routing IoT / Hazard Fusion]
    G --> H[IOT Sensor Telemetry future]
    G --> I[user_reports]

    subgraph Supabase
        B1[(user_reports)]
        B2[(sos_dispatches)]
        B3[(sos_dispatch_offers)]
        B4[(profiles)]
        B5[RPCs + Triggers]
    end
end

```

4) Core Logic Diagram

Flood report + duplicate gate + smart routing update

```

flowchart TD
    U[User opens Reports page] --> L[Get GPS]
    L --> R[Resolve address/landmark]
    R --> N[Load nearby flood reports]
    N --> M[Show mini-map + nearby markers]
    M --> S[User clicks Submit]
    S --> D[Duplicate check RPC]
    D -->|Duplicate found| W[Show warning + block submit]
    D -->|No duplicate| I[Insert into user_reports]
    I --> H[Hazard dataset updated]
    H --> SR[Smart route recompute for active navigations]

```

SOS lifecycle

```
flowchart TD
    P[User triggers SOS] --> Q[Create sos_dispatches row]
    Q --> S1[Status: submitted]
    S1 --> O[Offer generated to rescuers]
    O --> A[Rescuer accepts]
    A --> S2[Status: received/dispatching]
    S2 --> C[Rescuer starts moving]
    C --> S3[Status: en_route]
    S3 --> E[Arrival]
    E --> S4[Status: closed]
```

5) End-to-End Workflow Flowchart

```
flowchart LR
    U1[Citizen]
    U2[Rescuer]
    APP[Flutter App]
    DB[(Supabase)]
    API[FastAPI Routing Service]
    IOT[IOT Sensors future]

    U1 --> APP
    APP --> DB
    APP --> API
    IOT --> API
    API --> APP
    DB --> U2
    U2 --> APP
    APP --> DB
```

6) Key Technical Contributions (Defense Points)

- **Separated workflows:** `EmergencyPage` dedicated for SOS; `IncidentReportPage` dedicated for flood reports.
- **Duplicate prevention:** client-side nearby preview + server-side RPC (`is\_duplicate\_flood\_report`) before insert.
- **Improved location quality:** geocode/landmark resolution + fallback label (`Near lat,lng`) when needed.
- **Data consistency:** normalized `admin\_decision` values and SQL constraints/triggers on `user\_reports` (where deployed).
- **Navigation safety:** hazard-aware routing and strict reroute acceptance for flood-safe results.
- **Smart Routing IoT positioning:** clear extension path from crowdsourced reports to fused IoT+crowd hazard scoring for routing.

7) Risks and Mitigations (Short)

- **Risk:** Geocode key restrictions fail and return generic labels.
- **Mitigation:** fallback to precise `Near lat,lng`; keep accurate coordinates.

- **Risk:** Duplicate spam in same flooded area.
- Mitigation:** RPC duplicate radius/time-window validation before insert.
- **Risk:** UI confusion between SOS and reporting.
- Mitigation:** separated pages and responsibilities.
- **Risk:** IoT noise or false readings (when sensors are added).
- Mitigation:** thresholding, smoothing, and confidence weighting before affecting routes.

8) Possible Thesis Defense Questions with Short Answers

Q: Why separate SOS and incident reporting pages?

A: To reduce UX confusion, isolate responsibilities, and simplify maintenance and testing.

Q: Why use both client and server duplicate checks?

A: Client gives immediate UX feedback; server is authoritative and tamper-resistant.

Q: Why keep lat/long even if address resolution fails?

A: Coordinates are the most reliable source for routing and emergency response.

Q: What is your duplicate definition?

A: A report within configured radius (e.g., 120 m) and recent time window (e.g., 3 hours) with active hazard-related statuses (`pending`, `impassable`, `risky`), as implemented in policy and RPC.

Q: How does this support rescuer operations?

A: Cleaner flood data improves hazard awareness on the map and routing, and avoids redundant low-quality reports.

Q: What happens without Google geocoding?

A: The system still works using GPS plus a coordinate-based fallback label.

Q: Why use a status lifecycle for SOS dispatch?

A: It enables auditable response tracking and operational visibility from `submitted` through `closed`.

Q: What does “Smart Routing IoT System” add beyond maps?

A: It describes the fusion of dynamic hazards (reports + future sensors) with automated route selection and safe rerouting, not only static display.

Q: Main scalability improvement you would do next?

A: Move geocode and heavy hazard fusion to the backend for secure keys, centralized retries, and caching.

9) Conclusion

Floote combines **SOS dispatch**, **structured flood reporting**, and a **Smart Routing IoT** vision—crowd and sensor data feeding hazard-aware routing—so the platform supports both immediate rescue and safer everyday navigation during floods.